PUBLIC RELATIONS

Presented By :
Seongwon Jang

Present To
:



1. 자기 소개



장성원

2000.11.23.

GitHub
http://github.com/JungSeong

HuggingFace
http://huggingface.co/JungSeong2

Dacon
http://dacon.io/myprofile/537952/home

LinkedIN
http://www.linkedin.com/in/seongwonjang

학력

26.02 | 세종대학교 학사 졸업
평점 4.02/4.5 (전공 평점 4.13/4.5)

자격/수상

2025.12 | 정보처리기사 **2020.11** | 자동차운전면허 2종 보통
2025.06 | ROS2와 AI를 활용한 자율주행&로봇팔 개발자 부트캠프 최우수상
2024.12 | 2024 IN-JEJU Challenge 해커톤 우수상

경력

25.07–09 | UNIST 연구 인턴십
- Mixture-of-Expert 모델의 효율적인 파라미터 파인튜닝 기법 연구 개발

주요 프로젝트

- AI for Industry Challenge
- RoboDine
- 구조물 안정성 물리 추론 AI 경진대회

학내외 활동

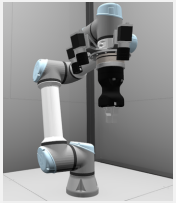
2026 | 모두의 연구소 Advanced Computer Vision Lab 스터디원
2025 | 오로카 판교 로봇 스터디원
2024 | 교내 프로그래밍 동아리 Prompt C언어 팀장

→

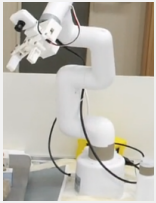
1. 자기 소개

기술 스택

HW



Universal Robotics (UR5e)

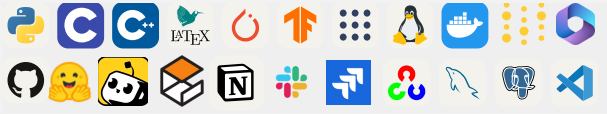


MyCobot280



Pinky

SW



교육

2025.04-06 | ROS2와 AI를 활용한 자율주행&로봇팔 직업 훈련

자격/면허

2026.03 | 정보처리기사

2020.10 | 2종 보통 면허

어학

2026.03 | 오픽 IH

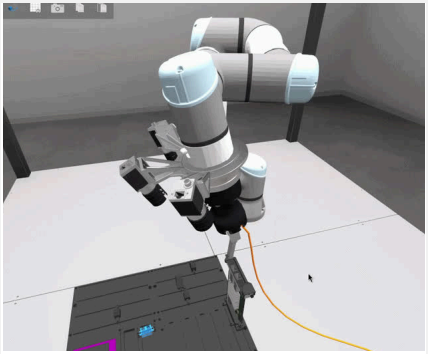
2025.06 | 토익 925

2. 주요 프로젝트

2-1. AI for Industry Challenge
2-2. RoboDine
2-3. 구조물 안정성 물리추론 AI 경진대회

2-1. AI for Industry Challenge

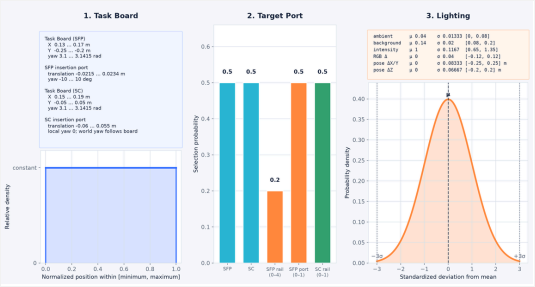
담당 역할	팀장
주요 업무	• Gazebo Simulator 기반 랜덤화 환경 및 정책별 데이터 수집 ROS2 노드 개발 • **YOLO Keypoint Detection 및 Multiview Triangulation** 기반 포트 3-DoF 추론 로직 구현 • Fine-Grained한 상황에서 케이블 정렬을 위한 **MultiView Bilinear Cross-Attention Regressor** 구현 • 팀원 구인 및 GitHub, HuggingFace, Notion 관리
기술 스택	HW : Universal Robotics UR5e SW : Python, PyTorch, Ubuntu 24.04, ROS2 Kilted Kaiju, Gazebo, Docker, YOLOv8, OpenCV GPU : DGX Spark, NVIDIA RTX 3060



Peg-In-Hole

2-1. AI for Industry Challenge

1. Gazebo Simulator에서 동작하는 **도메인 랜덤화 데이터 세트 생성 및 수집 ROS2 노드 구현**
• Uniform Distribution으로 Task Board의 XY/yaw, 포트 타입 및 XY/yaw 변화, Gaussian Distribution으로 조명 변화를 둔 시나리오 생성과 HuggingFace Hub 업로드 로직 구현



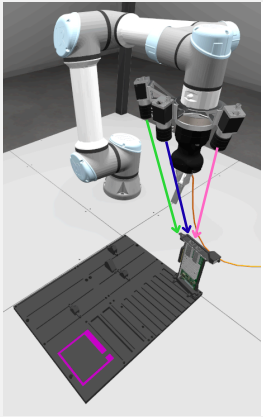
Distribution (Task Board, Port, Lightening)

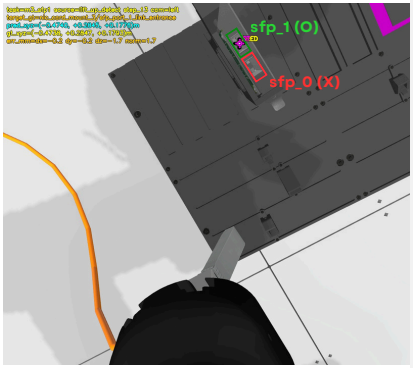


Data Collector

2-1. AI for Industry Challenge

2. **YOLO Keypoint Detection 및 Multiview Triangulation 기반 포트 3-DoF 추론 로직 구현**, 5개 Case에서 MAE 약 5mm 내외로 예측 정확도 달성



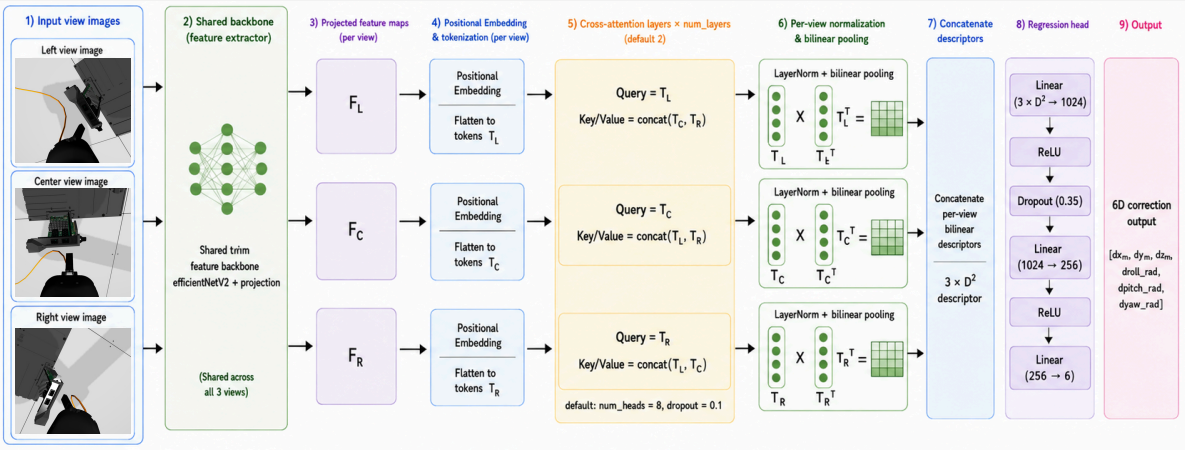


	Type	Ground Truth	Estimated
1	card0/sfp_0	(-0.4442, +0.2118, +0.1793)	(-0.4452, +0.2107, +0.1767)
2	card1/sfp_1	(-0.4435, +0.2535, +0.1793)	(-0.4441, +0.2521, +0.1675)
3	card2/sfp_0	(-0.3843, +0.3501, +0.1793)	(-0.3840, +0.3507, +0.1737)
4	card3/sfp_1	(-0.4738, +0.2847, +0.1793)	(-0.4740, +0.2845, +0.1776)
5	card4/sfp_0	(-0.4248, +0.3330, +0.1793)	(-0.4701, +0.3406, +0.1825)

$$\mathrm{MAE} = \frac{1}{5 \times 3} \sum_{i=1}^{5} \left(|dx_i| + |dy_i| + |dz_i| \right)$$
$$= \boxed{5.536 \text{ mm}}$$

3. Fine-Grained한 상황에서 케이블 정렬을 위한 **MultiView Bilinear Cross-Attention Regressor** 구현

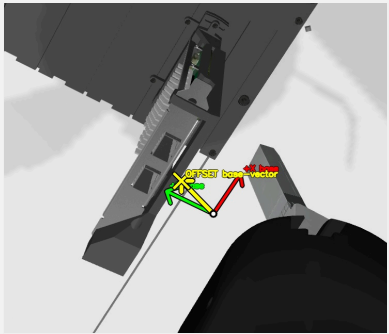
- **Worst Case 대비 XYZ MAE 21.3% 및 RPY MAE 16.5% 향상**

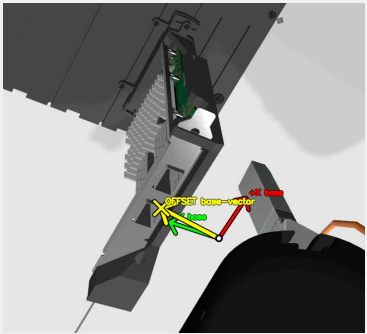


2-1. AI for Industry Challenge

3. Fine-Grained한 상황에서 케이블 정렬을 위한 **MultiView Bilinear Cross-Attention Regressor** 구현

- **Case에서 Worst Case 대비 XYZ MAE 21.3% 및 RPY MAE 16.5% 향상**





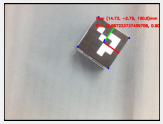
	Type	model	xyz_mae (mm)	rpy_mae (deg)
1	SFP	cross_attention_bilinear **(Best)**	9.251	2.207
2	SC	cross_attention_bilinear **(Best)**	9.298 (-21.3%)	2.541 (-16.5%)
3	SFP	shared_bilinear	9.452	2.485
4	SFP	multiview_bilinear	9.536	2.620
5	SC	shared_bilinear	9.707	2.805
6	SFP	simple_cnn	9.985	2.538
7	SC	simple_cnn	10.56	2.791
8	SC	multiview_bilinear **(Worst)**	11.82	3.043

2-2. RoboDine

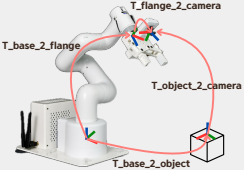
담당 역할	팀원
주요 업무	• 데이터 전처리 및 파라미터 튜닝을 통한 단안 2D 카메라가 달린 로봇팔의 객체 3-DoF 추정 정확도 14% 향상 • 단안 카메라 및 2D LiDAR 거리 정보를 융합한 ROS2 기반 장애물 인식 및 회피 노드 설계
기술 스택	HW : MyCobot280, Pinky SW : Python, PyTorch, Ubuntu 24.04, ROS2 Jazzy, OpenCV, YOLOv8 GPU : NVIDIA RTX 3060

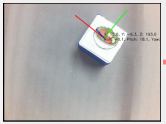
2-2. RoboDine

1. **데이터 전처리 및 파라미터 튜닝을 통한 단안 2D 카메라가 달린 로봇팔의 객체 3-DoF 추론 정확도 14% 향상**
   - 데이터 증강, Dropout, Pose Loss 계수 상향해 YOLO pose 모델 학습
   - 카메라 캘리브레이션 내부 파라미터와 YOLO 추론값으로 solvePnP 연산

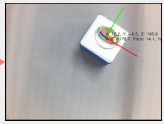


ARUCO





BEFORE
(추정이 잘못 됨)

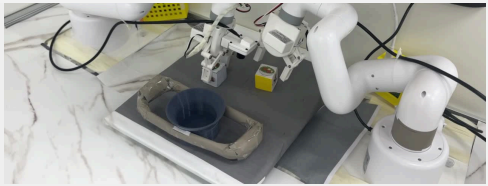


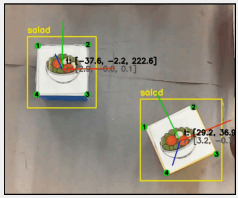
AFTER
(추정이 잘 됨)

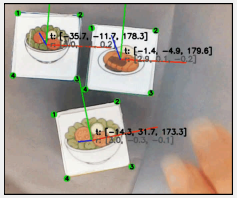
vs ARUCO	초기 모델	최종 모델
1	10.0189	3.188
2	5.2291	7.9605
3	5.3206	12.2327
4	20.8498	9.8768
5	21.9831	13.4475
6	4.3244	6.4558
7	4.9146	5.8002
8	8.6163	12.7653
9	1.7042	0.6268
10	6.9252	5.2689
average error	8.9886	7.7622

-14%

*다의: mm



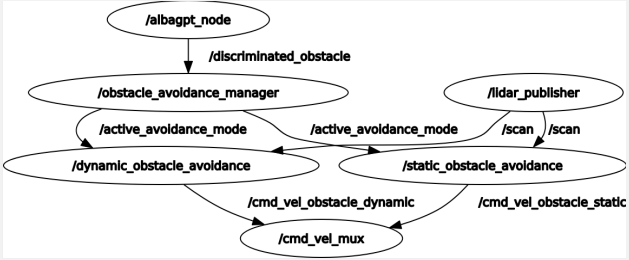


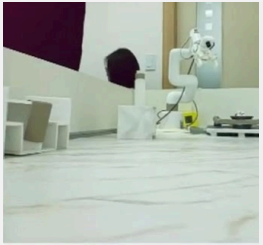


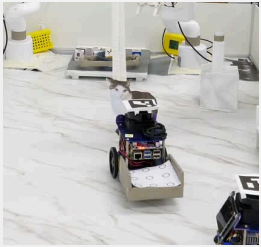
Pose Estimation

2. ROS2 기반 단안 카메라 및 2D LiDAR 거리 정보를 융합한 장애물 인식 및 회피 노드 설계

- **라이다 거리 30CM 이내 + 동적 장애물**: 정지 이후 장애물이 시야에서 사라지면 이동
- **라이다 거리 20CM 이내 + 정적 장애물**: 회피 이후 이동

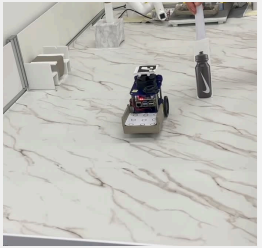






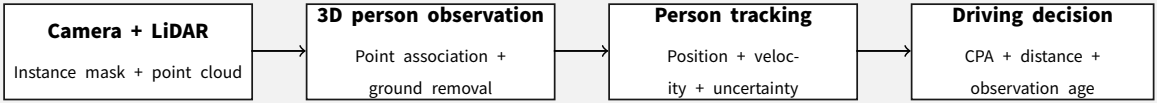
Dynamic Obstacle

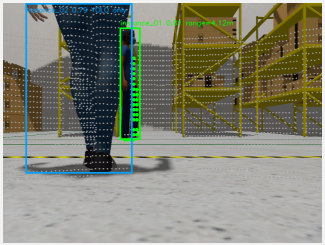




Mobin

From sensors to person tracking and driving decisions





Observation processing

Use TF at the point-cloud timestamp; associate person points after ground removal.

Tracking and decision

Update odom-frame tracks with Hungarian assignment; pass position and velocity to the decision logic.

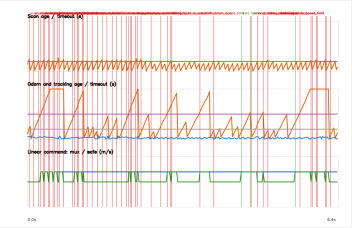
Existing capture: YOLO-only and distance-associated observations; not every detection has a valid distance.

Mobin

Execution results and remaining observation and tracking limits

Existing 60-second seeded-multi-person-tracking run. This is a single-run result, not a full performance statistic.

Metric	Recorded value
Common sensor ROI coverage	16.67%
IDF1	0.0595
ID switches	2
Ground-truth misses (FN)	130
Patrol state	not_started



Interpretation

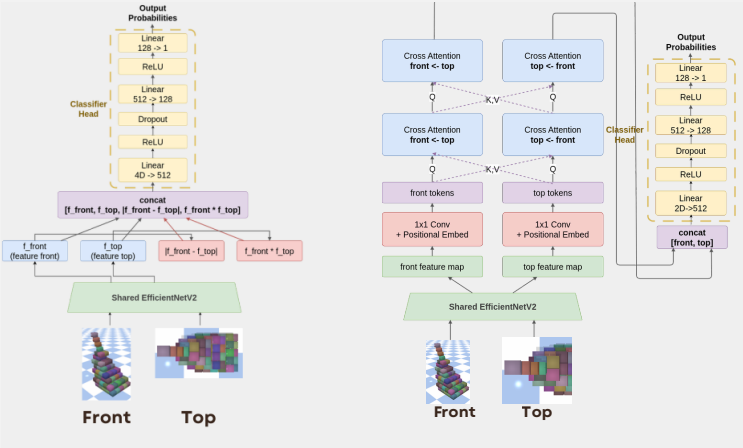
Common observation coverage and tracking quality remain limited. The run also recorded safety behavior triggered by observation age.

Values are taken from summary.json. They do not establish completed root-cause analysis or safe autonomous driving.

2-3. 구조물 안정성 물리 추론 AI 경진대회

담당 역할	팀원
주요 업무	• Multi-View Bidirectional Cross Attention 모델 설계 및 적용해 기존 Feature Fusion 모델 대비 logloss 48.2% 및 Accuracy 3.13% 개선 • VideoMAE로 영상 특징 벡터 추출 및 Distillation Regularization 추가해 Baseline 대비 logloss 86.3% 및 accuracy 9% 개선 • Backbone, Epochs, Batch Size 등 HyperParameter Search 수행
기술 스택	SW : Python, PyTorch, Ubuntu 24.04 GPU : DGX Spark

1. **Multi-View Bidirectional Cross Attention** 모델 설계 및 적용
- 기존 Feature Fusion 모델 대비 **logloss 48.2% 및 Accuracy 3.13% 개선**



모델\val	logloss (↓)	accuracy (↑)	AUC (↑)
Feature Fusion **(Baseline)**	0.1216	0.96	0.9927
Cross Attention	0.2323	0.89	0.9743
Bidirectional Cross Attention **(Ours)**	0.063 (-48.2%)	0.99 (+3.13%)	0.9983 (+0.56%)

2-3. 구조물 안정성 물리 추론 AI 경진대회

2. VideoMAE로 영상에서 특징 벡터 추출 및 Distillation Regularization 추가해 학습

• Baseline 대비 **logloss 86.3% 및 accuracy 9% 개선**

$$\left(\begin{array}{c} \mathcal{L}_{\text{video_reg}} = \dfrac{1}{|\mathcal{M}| \cdot 768} \displaystyle\sum_{i \in \mathcal{M}} \left\| z_i^{\text{img}} - z_i^{\text{video}} \right\|_2^2 \\ \mathcal{L}_{\text{total}} = \mathcal{L}_{\text{BCE}} + \lambda \mathcal{L}_{\text{video_reg}} \end{array} \right)$$

모델	Metrics (↓)	Epochs (↓)	logloss (↓)	accuracy (↑)
efficientnet v2_rw_s **(Baseline)**	-	100	0.218147	0.91
efficientnet v2_rw_s **(BEST)**	Distillation	30	0.02986 (-86.3%)	1.00 (+9%)
efficientnet v2_rw_s	Distillation	50	0.09056	0.98
efficientnet v2_rw_s	Distillation	100	0.05361	0.99

3. Backbone, Epochs, Batch Size 등 HyperParameter Search 수행

• **Worst Trial 대비 logloss 95.5% 및 accuracy 40% 개선**

※ CosineAnnealingLR 스케줄러 사용, Epochs에 따라서도 수렴 속도

1	efficientnet v2_rw_s **(BEST)**	30	0.02986 (-95.5%)	1.00 (+40%)
2	efficientnet v2_rw_s	50	0.090561	0.98
3	convnext tiny	30	0.113118	0.97
4	convnext tiny	50	0.322998	0.81
5	swin tiny	30	0.222444	0.93
6	swin tiny **(WORST)**	50	0.657001	0.60